# Pattern-Oriented Modeling

## EES 4760/5760

## Agent-Based and Individual-Based Computational Modeling

J. Magnolia Gilligan

Class #20: Thursday, November 5, 2026

# Breaking News

# Abdul El-Sayed

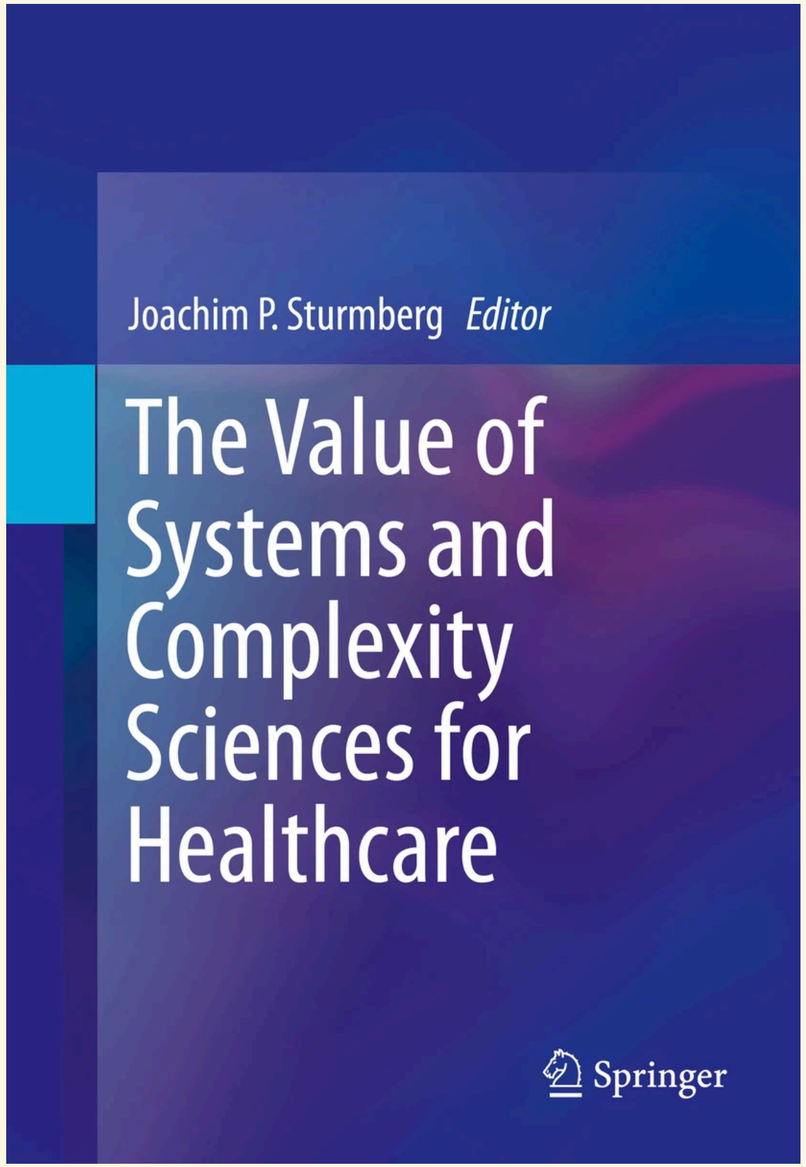
## Dr. Abdul El-Sayed:

- Ph.D. in Public Health, U. Oxford (2011)
- M.D., Columbia U. (2014)
- Assistant Professor of Epidemiology, Columbia U. 2014–2015
  - Director Systems Science Program & Global Research Analytics for Population Health
- Executive Director, Detroit Health Dept. 2015–2022
- Director, Wayne County MI Dept. of Health, Human, & Veterans' Services 2023–2025
- Elected U.S. Senator from Michigan, 2026
- More than 100 publications in public health research
  - His work has been cited more than 4,000 times



Photo: Scott Olson/Getty Images

# El-Sayed and Agent-Based Modeling




**Chapter 4**

## A System for Systems Epidemiology: The Example of Inference from Agent-Based Models

**Abdulrahman M. El-Sayed**

### 4.1 Systems and Epidemiology: The Conceptual Argument

The dynamics of populations, in terms of health and disease, emerge from the behaviors and interactions of the heterogeneous individuals that comprise them. In this way, interaction undergirds many of the mechanisms that mediate the production of health and disease. These interactions may operate both on the macro-scale between exposures acting at multiple levels and on the micro-scale between individuals within populations. Interactions challenge the current epidemiologic toolkit in several ways. As factors may interact in complex ways to determine health and disease risk, the current "risk factor" approach to epidemiology, which emphasizes decontextualized, independent effect measures for exposures may not be appropriate [3, 4]. Furthermore, variability in health may be mediated by the degree and nature of social interaction within and between social groups. In this regard, several studies have shown that social interaction may transmit non-infectious disease outcomes [5–8]. Furthermore, research about the health influences of social

A.M. El-Sayed, "A System for Systems Epidemiology: The Example of Inference from Agent-Based Models", in J.P. Sturmberg (ed) *The Value of Systems and Complexity Sciences for Healthcare*, pp. 39–49 (Springer, 2016) doi: 10.1007%2F978-3-319-26221-5_4

# El-Sayed and Agent-Based Modeling

El-Sayed *et al. Epidemiologic Perspectives & Innovations* 2012, **9**:1
http://www.epi-perspectives.com/content/9/1/1



EPIDEMIOLOGIC PERSPECTIVES & INNOVATIONS

**ANALYTIC PERSPECTIVE** **Open Access**

## Social network analysis and agent-based modeling in social epidemiology

Abdulrahman M El-Sayed[1,2,3*], Peter Scarborough[1], Lars Seemann[4] and Sandro Galea[2]

**Abstract**

The past five years have seen a growth in the interest in systems approaches in epidemiologic research. These approaches may be particularly appropriate for social epidemiology. Social network analysis and agent-based models (ABMs) are two approaches that have been used in the epidemiologic literature. Social network analysis involves the characterization of social networks to yield inference about how network structures may influence risk exposures among those in the network. ABMs can promote population-level inference from explicitly programmed, micro-level rules in simulated populations over time and space. In this paper, we discuss the implementation of these models in social epidemiologic research, highlighting the strengths and weaknesses of each approach. Network analysis may be ideal for understanding social contagion, as well as the influences of social interaction on population health. However, network analysis requires network data, which may sacrifice generalizability, and causal inference from current network analytic methods is limited. ABMs are uniquely suited for the assessment of health determinants at multiple levels of influence that may couple with social interaction to produce population health. ABMs allow for the exploration of feedback and reciprocity between exposures and outcomes in the etiology of complex diseases. They may also provide the opportunity for counterfactual simulation. However, appropriate implementation of ABMs requires a balance between mechanistic rigor and model parsimony, and the precision of output from complex models is limited. Social network and agent-based approaches are promising in social epidemiology, but continued development of each approach is needed.

**Keywords:** Complex systems, epidemiology, social factors, social class, ethnicity, modeling, networks

A.M. El-Sayed, P. Scarborough, L. Seemann, & S. Galea, "Social Network Analysis and Agent-Based Modeling in Social Epidemiology," *Epidemiologic Perspectives & Innovations* **9**, 1 (2012) doi: 10.1186/1742-5573-9-1.



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Vol. 178, No. 2
DOI: 10.1093/aje/kws455
Advance Access publication:
May 21, 2013

**Practice of Epidemiology**

## Are Network-Based Interventions a Useful Antiobesity Strategy? An Application of Simulation Models for Causal Inference in Epidemiology

**Abdulrahman M. El-Sayed*, Lars Seemann, Peter Scarborough, and Sandro Galea**

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*Initially submitted April 4, 2012; accepted for publication November 12, 2012.*

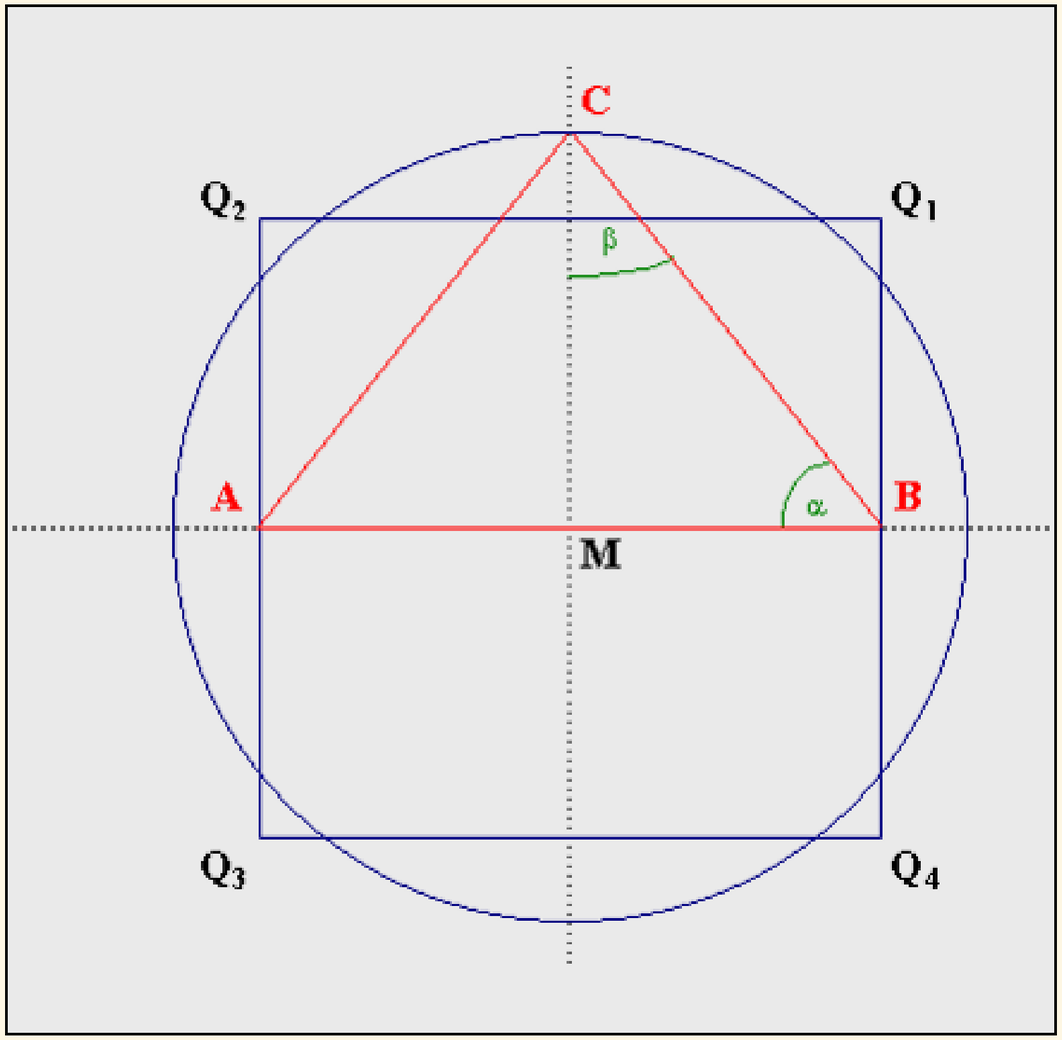
Recent research suggests that social networks may present an avenue for intervention against obesity. By using a simulation model in which artificial individuals were nested in a social network, we assessed whether interventions targeting highly networked individuals could help reduce population obesity. We compared the effects of targeting antiobesity interventions at the most connected individuals in a network with those targeting individuals at random. We tested 2 interventions, the first "preventing" obesity among 10% of the population at simulation outset and the second "treating" obesity among 10% of the obese population yearly, each in 2 separate simulations. One simulation featured a literature-based parameter for the network spread of obesity, and the other featured an artificially high parameter. Interventions that targeted highly networked individuals did not outperform at-random interventions in simulations featuring the literature-based parameter. However, in simulations featuring the artificially high parameter, the targeted prevention intervention outperformed the at-random intervention, whereas the treatment intervention implemented at random outperformed the targeted treatment intervention. Results were qualitatively similar across network topologies and intervention scales. Although descriptive studies suggest that social networks influence the spread of obesity, policies targeting well-connected individuals in social networks may not improve obesity reduction. We highlight and discuss the potential applications of counterfactual simulations in epidemiology.

computer simulation; intervention; obesity; public health; social networks

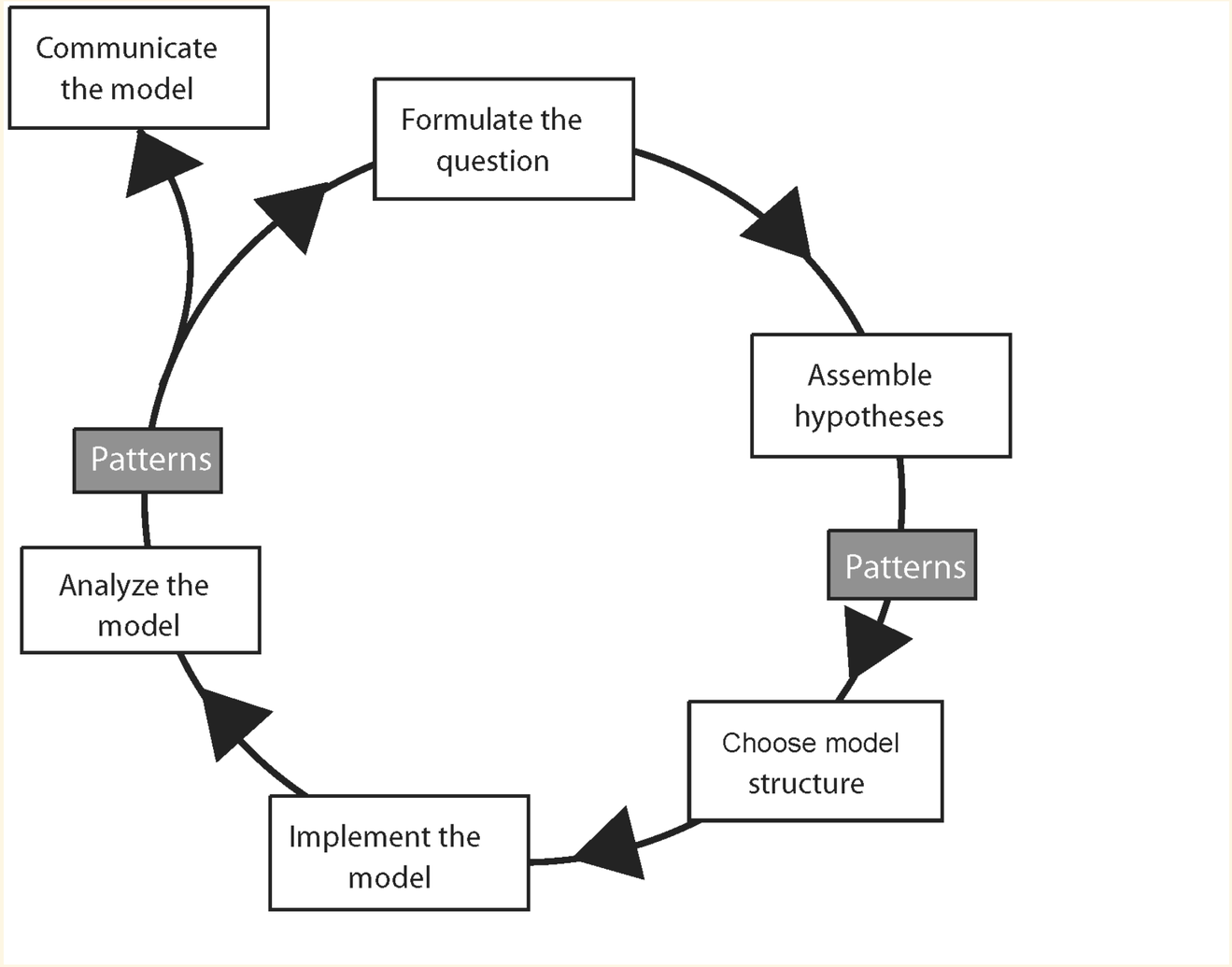
A.M. El-Sayed, P. Scarborough, L. Seemann, & S. Galea, "Are Network-Based Interventions a Useful Antiobesity Strategy? An Application of Simulation Models for Causal Inference in Epidemiology," *Am. J. Epidemiology* **178**, 287–95 (2013). doi: 10.1093/aje/kws455.

# Pattern-Oriented Modeling

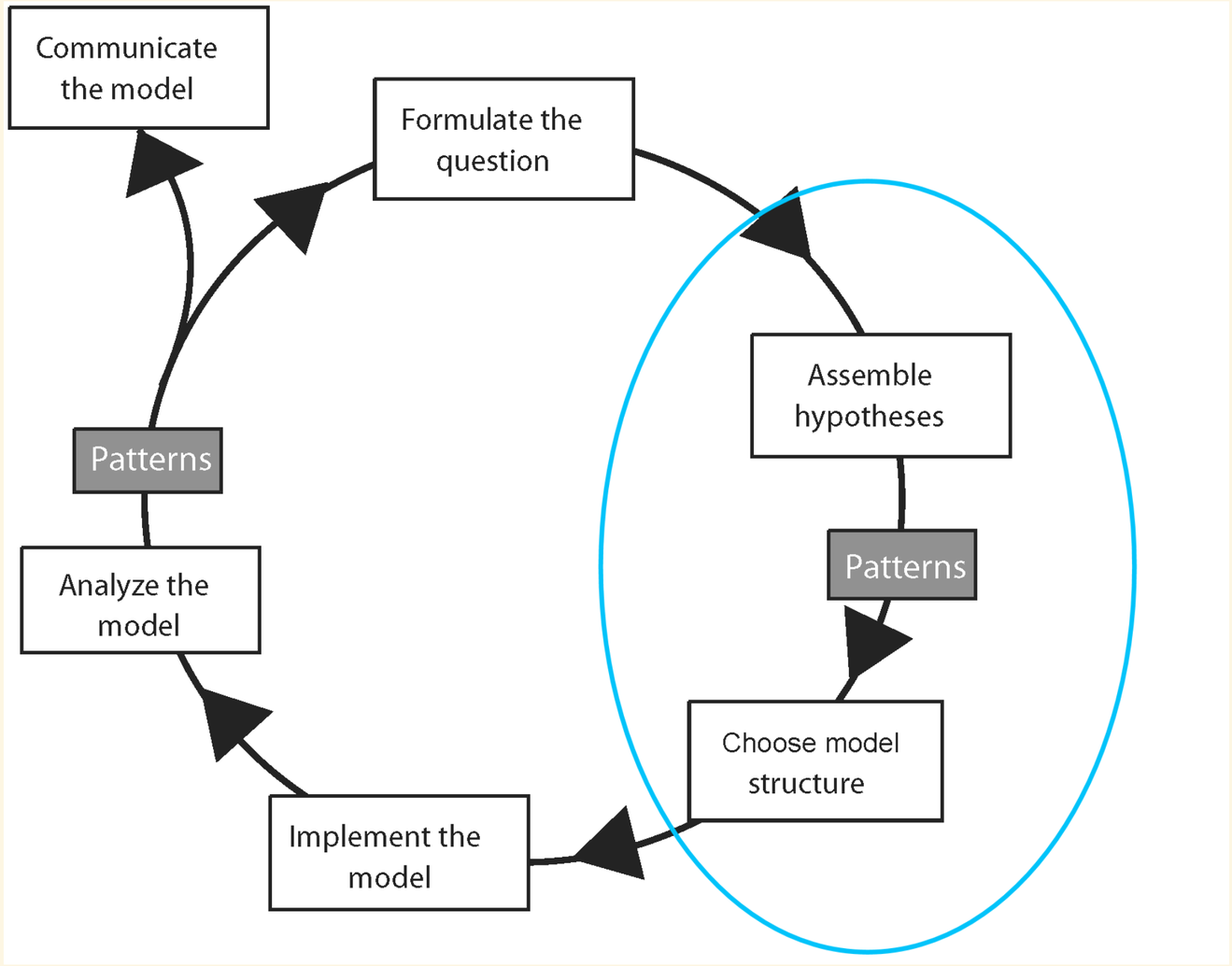
# Pattern-Oriented Modeling



# The Modeling Cycle



# The Modeling Cycle



# Validation and Verification

# Goals of Modeling

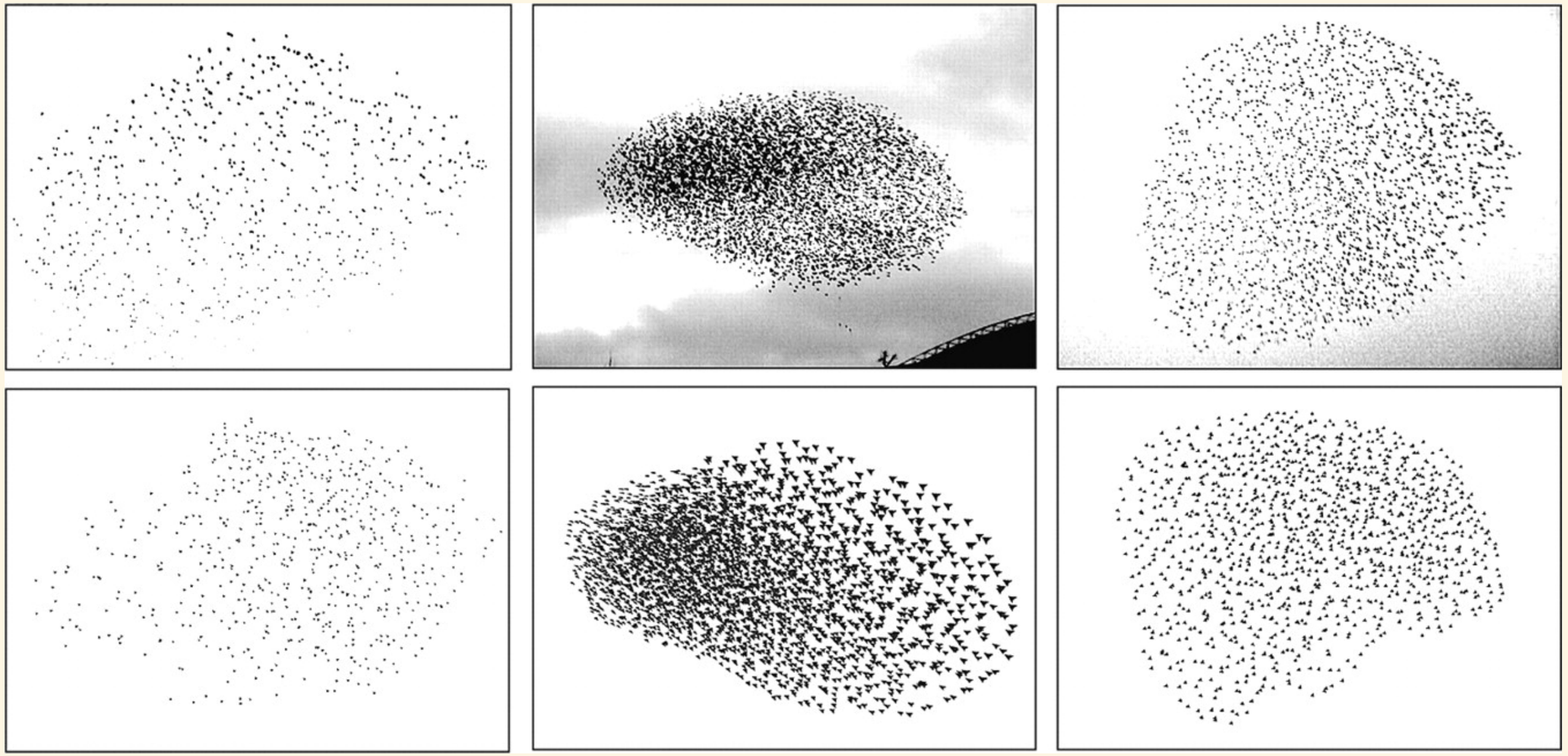
- Goals:
  - Models are “sufficiently good” representations of real counterparts
  - Learn about real world:
    - Capture essential elements of real system’s *internal organization*
    - Capture *generative mechanisms* that produce structure and behavior of real systems

# Problem: Validation and Verification

- **Version 1:**
  - The model *mimics the real world well enough for its stated purpose*
- **Version 2**
  - We can place confidence in *inferences about real system based on model results*

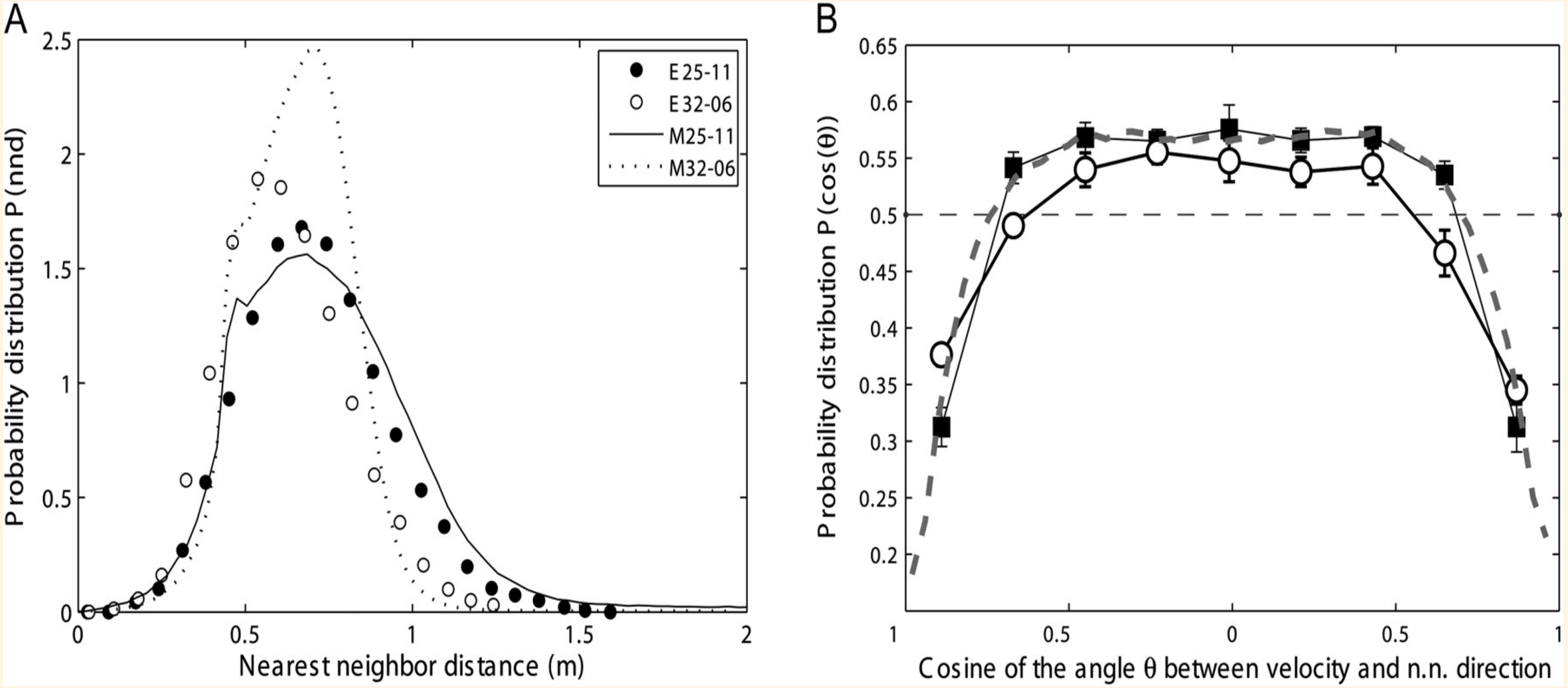
E.J. Rykiel “Testing ecological models: The meaning of validation” Ecological Modeling **90**, 229 (1996)

# Starling Flocks



H. Hildenbrandt *et al.*, “Self-organized aerial displays of thousands of starlings: a model,” Behav. Ecol. **21**, 1349–1359 (2010).

# Validation of Starling Model




H. Hildenbrandt *et al.*, "Self-organized aerial displays of thousands of starlings: a model," Behav. Ecol. **21**, 1349–1359 (2010).

# Fundamental Problem

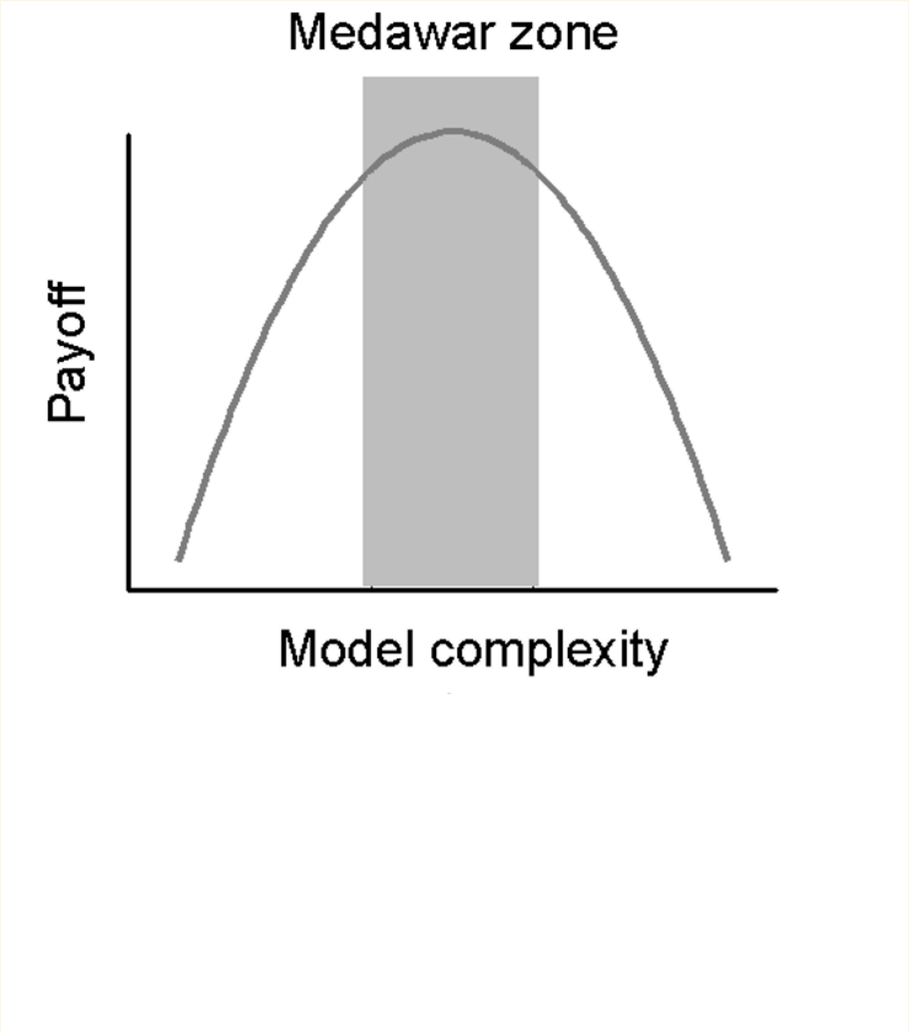
- Our model might reproduce the *right pattern* for the *wrong reasons*
- How can we be sure to capture the *real generative mechanisms*?
- How can we design models so that we can *optimize* model complexity?

# The Medawar Zone

**P. Medawar (1967):**

Optimal level of difficulty for a good research problem:

- Not difficult enough:
  - Result is trivial, uninteresting.
- Too difficult:
  - Unlikely to solve it




P. Medawar, *The Art of the Soluble* (Oxford, 1967)

# Mechanistically Rich Models

- If model structure is too simple it will not capture essential mechanisms
  - There will be too few ways to test the model
- Complexity of model is not bad *per se* and can increase the payoff

D.L. De Angelis and W.M. Mooij, “In Praise of Mechanistically Rich Models,” in C.D. Canham, J.J. Cole, and W.K. Lauenroth (eds.), *Models in Ecosystem Science* (Princeton, 2003)

# Spatial Patterns in Ecology

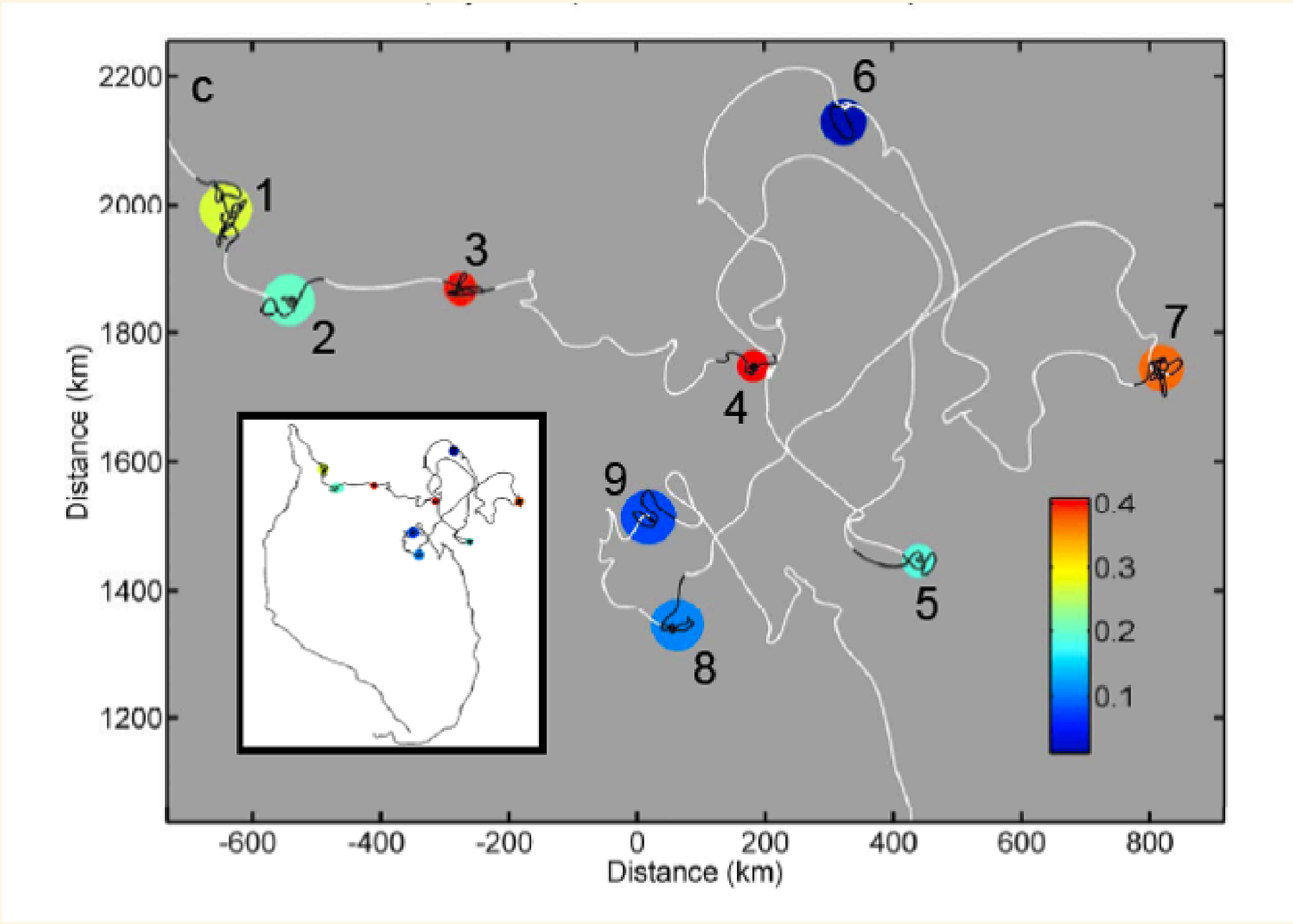
# Spatial Patterns in Ecology



Wave forest of *Empetrum nigrum* (Crowberry), Mistaken Point Ecological Reserve, Newfoundland. Photo: John Maunder.

http://www.digitalnaturalhistory.com/images/empetrumnigrumlivedeadwaveforestmistakenpoint.jpg

# Spatial Patterns in Ecology




M. Antolos, S.A. Shaffer, H. Weimerskirch, Y. Tremblay, and D.P. Costa, “Foraging Behavior and Energetics of Albatrosses in Contrasting Breeding Environments,” Front. Mar. Sci. **4**, 414 (2017). doi: 10.3389/fmars.2017.00414

# What Scientists do with Patterns

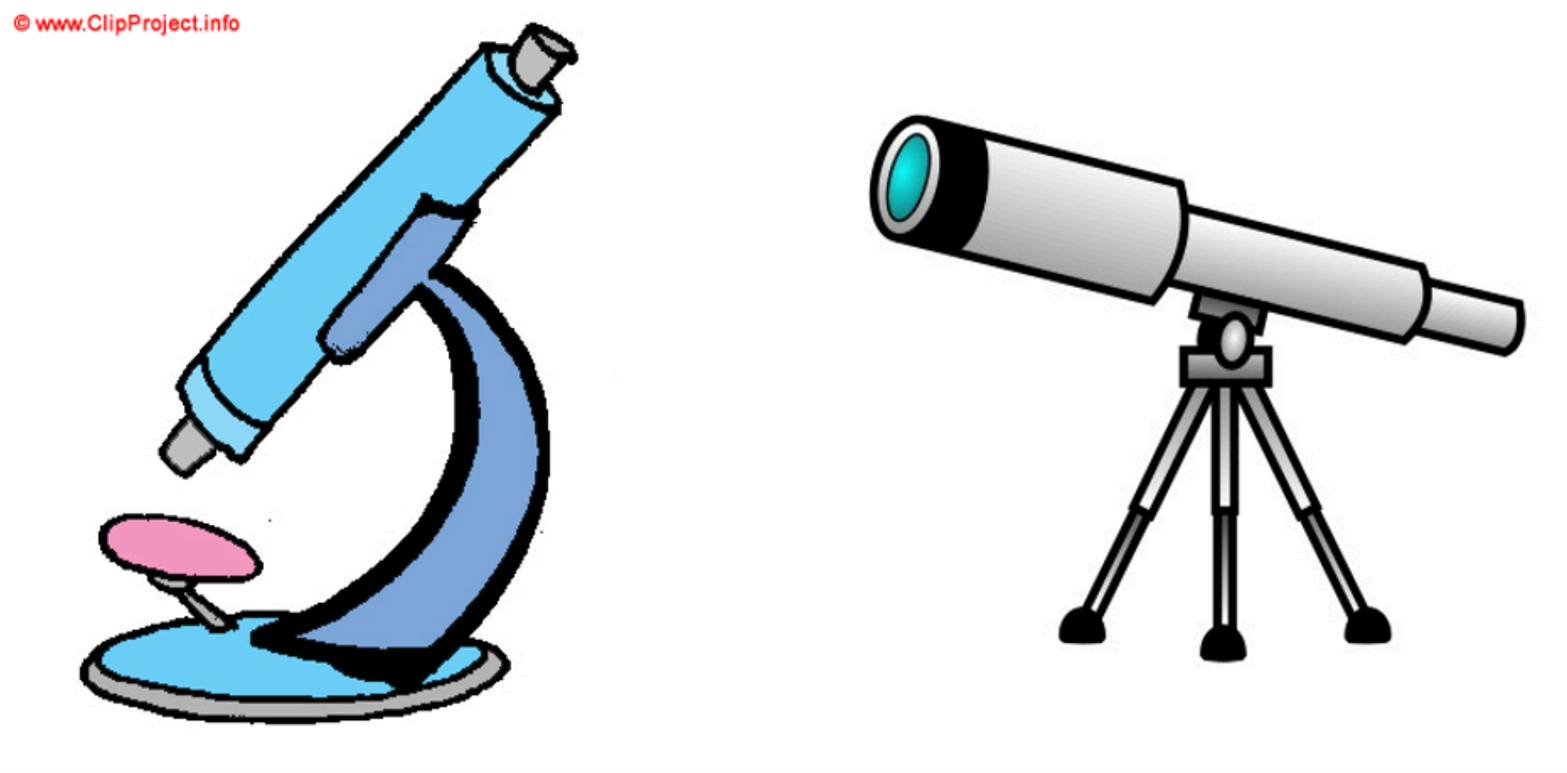



- **Pattern:** Something beyond random variation
- Pattern contains information about internal organization
- Develop models that reproduce observed pattern
- Inference: Real system’s internal mechanisms are like model’s
- ***Squeeze*** the pattern!

# Complex Systems

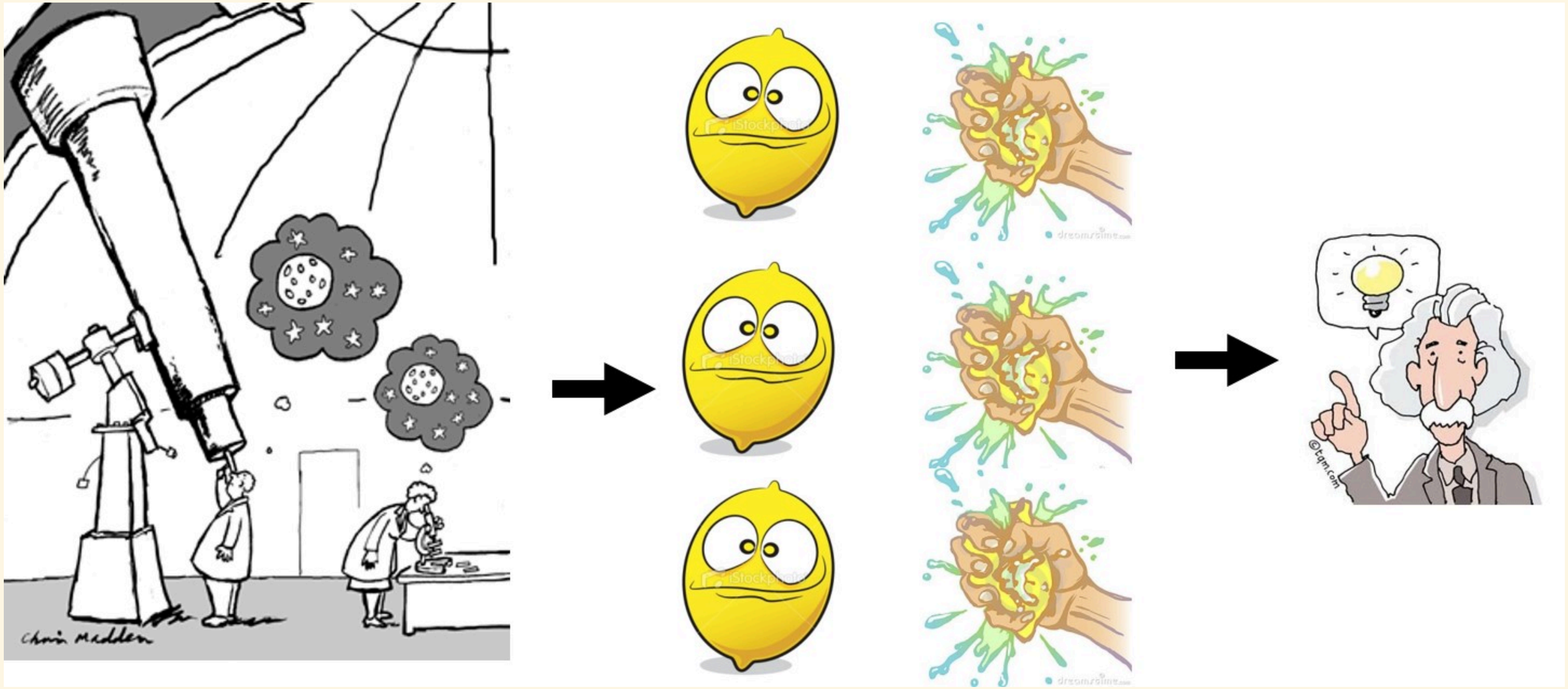
- A single pattern may not contain enough information
- Tendency to focus on *single* patterns observed at *one level* of observation:
  - Individual Behaviors
  - Population dynamics
  - Community composition

# Monoscopic View




Most approaches (and modelers) are not making the best use of available information (lemons)

# We Need a Multiscope



# Multiscope View

- Consider multiple patterns
  - Observed at different scales, levels of organization
- Get model to reproduce multiple patterns simultaneously
- Use each pattern as a *filter* to reject faulty submodels or parameterizations
- Multiple (3 or more) *weak* patterns may constrain model better than single *strong* pattern
- ***Pattern-Oriented Modeling***

# Pattern-Oriented Modeling

REVIEW

## Pattern-Oriented Modeling of Agent-Based Complex Systems: Lessons from Ecology

Volker Grimm,[1*] Eloy Revilla,[2] Uta Berger,[3] Florian Jeltsch,[4] Wolf M. Mooij,[5] Steven F. Railsback,[6] Hans-Hermann Thulke,[1] Jacob Weiner,[7] Thorsten Wiegand,[1] Donald L. DeAngelis[8]

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY B

*Phil. Trans. R. Soc. B* (2012) **367**, 298–310
doi:10.1098/rstb.2011.0180

*Research*

## Pattern-oriented modelling: a 'multi-scope' for predictive systems ecology

**Volker Grimm[1,*] and Steven F. Railsback[2]**

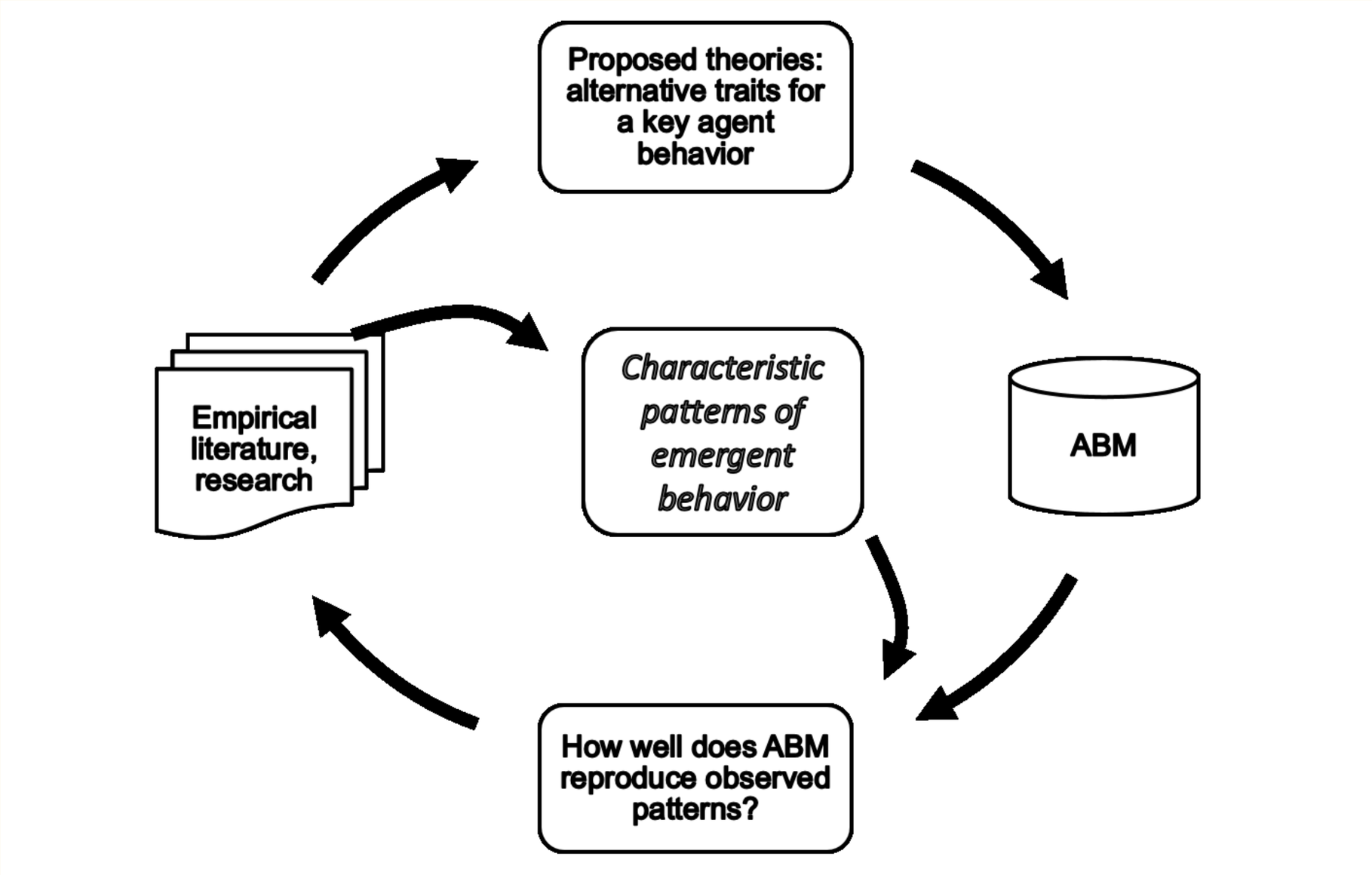
[1]*Department of Ecological Modelling, Helmholtz Centre for Environmental Research – UFZ, Permoserstrasse 15, 04318 Leipzig, Germany*
[2]*Department of Mathematics, Humboldt State University and Lang, Railsback and Associates, 250 California Avenue, Arcata CA 95521, USA*

Modern ecology recognizes that modelling systems across scales and at multiple levels—especially to link population and ecosystem dynamics to individual adaptive behaviour—is essential for making the science predictive. 'Pattern-oriented modelling' (POM) is a strategy for doing just this. POM is the multi-criteria design, selection and calibration of models of complex systems. POM starts with identifying a set of patterns observed at multiple scales and levels that characterize a system with respect to the particular problem being modelled; a model from which the patterns emerge should contain the right mechanisms to address the problem. These patterns are then used to (i) determine what scales, entities, variables and processes the model needs, (ii) test and select submodels to represent key low-level processes such as adaptive behaviour, and (iii) find useful parameter values during calibration. Patterns are already often used in these ways, but a mini-review of applications of POM confirms that making the selection and use of patterns more explicit and rigorous can facilitate the development of models with the right level of complexity to understand ecological systems and predict their response to novel conditions.

# Three Elements of Pattern-Oriented Modeling

1. **Design:** Choose state variables that allow real-world patterns to emerge in models.
2. **Selection:** Use multiple patterns to compare & reject submodels
3. **Parameterization:** Use multiple patterns to constrain entire sets of unknown parameters (*inverse modeling*)

# From Theories to Models and Back Again



# Example: Vultures and Carcasses

# Vultures Feeding at a Carcass



# Modeling Vulture Feeding: Roles

| Role | Description |
|---|---|
| Searcher | A vulture without personal or social information about carcass location |
| Finder | A vulture that has seen either the carcass, feeders on a carcass, or finders sinking in vertical flight to a carcass |
| Follower | A vulture that is following other vultures (either finders or other followers) |
| Feeder | A finder that has already arrived at a carcass |

# Modeling Vulture Feeding Behavior: Variables

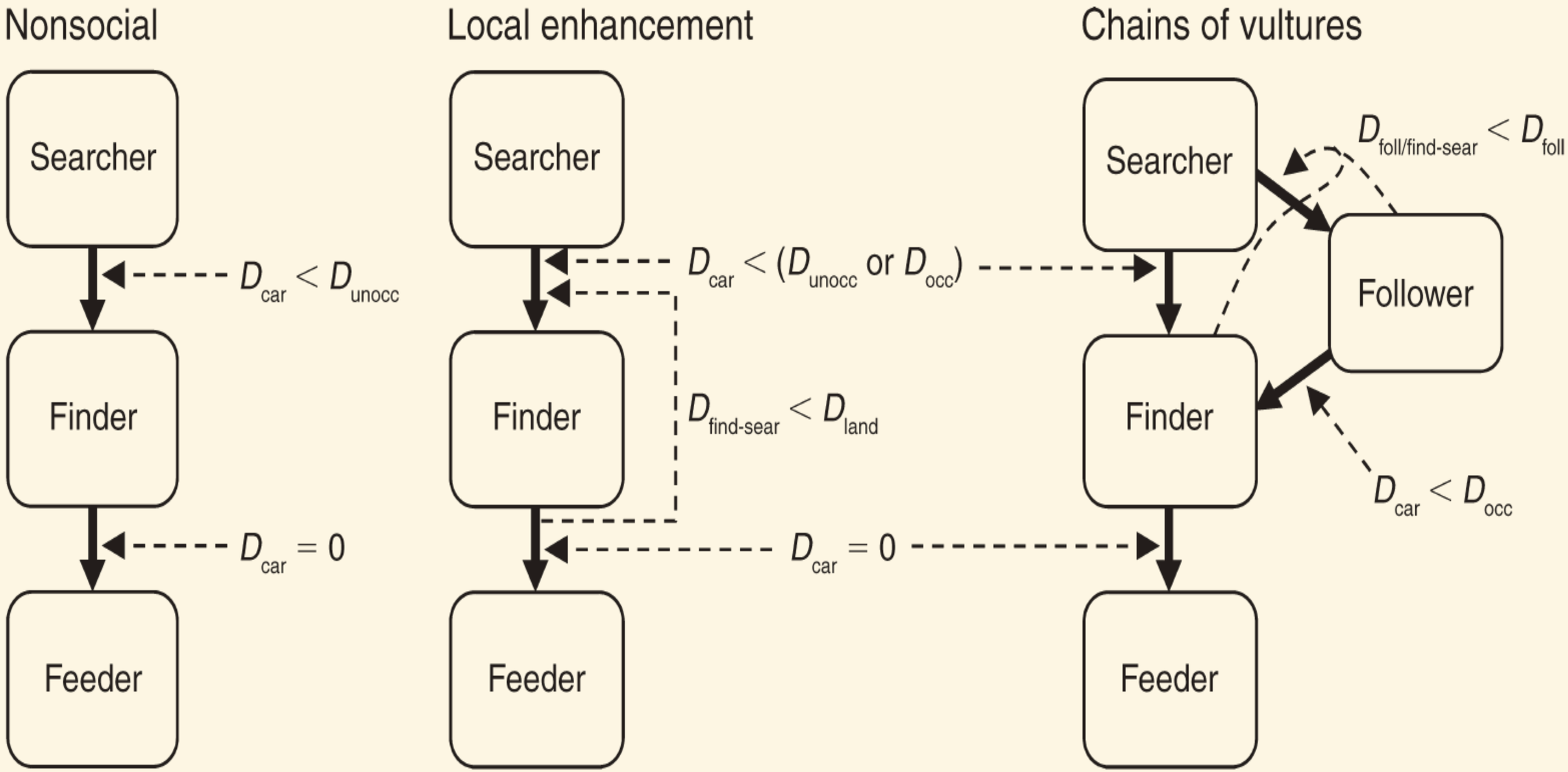
## Variables

| Variable | Meaning |
|---|---|
| $D_{car}$ | Distance to nearest carcass |
| $D_{find\text{-}sear}$ | Distance to a finder that’s landing to feed |
| $D_{foll/find\text{-}sear}$ | Distance to the nearest follower or finder |

## Fixed Parameters

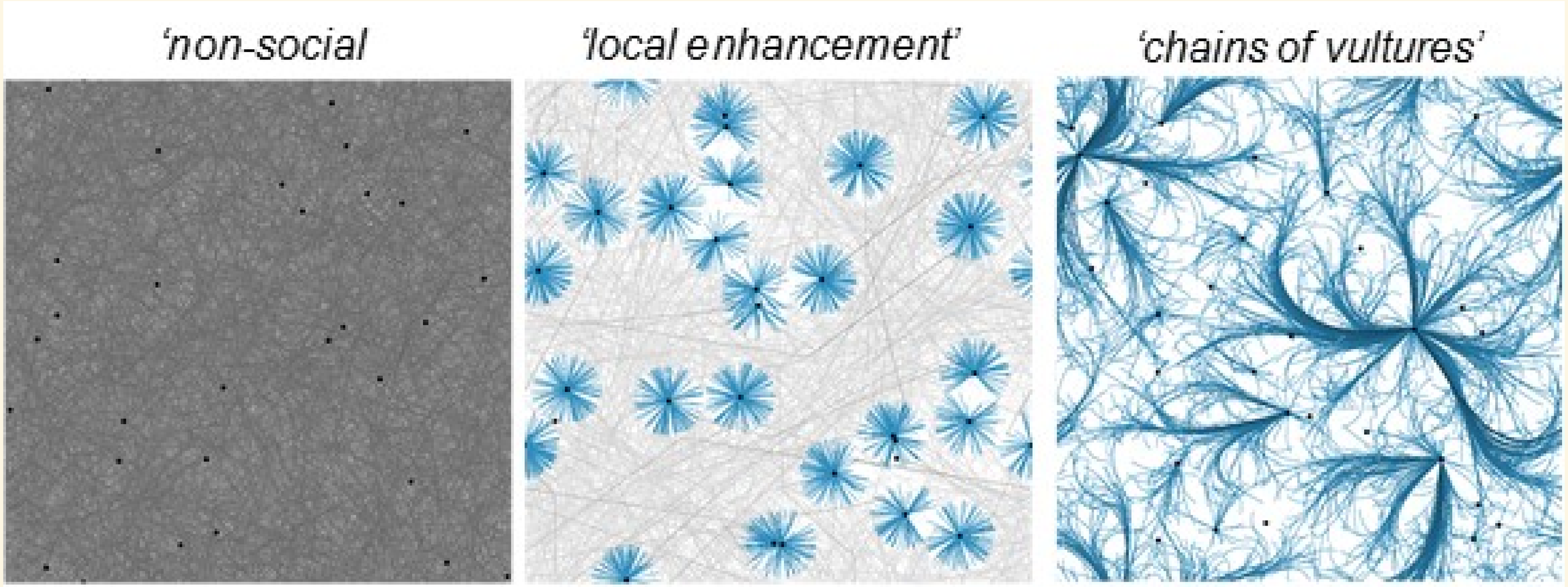
| Parameter | Meaning |
|---|---|
| $D_{unocc}$ | Farthest distance for vulture to see an unoccupied carcass |
| $D_{occ}$ | Farthest distance for vulture to see an occupied carcass, $D_{occ} > D_{unocc}$ |
| $D_{foll}$ | Farthest distance for vulture to see a finder or follower flying toward a carcass, $D_{foll} > D_{occ}$ |
| $D_{land}$ | Farthest distance for vulture to see a finder/follower landing on a carcass, $D_{land} = D_{foll}$ |

# Modeling Vulture Feeding Behavior: Submodels

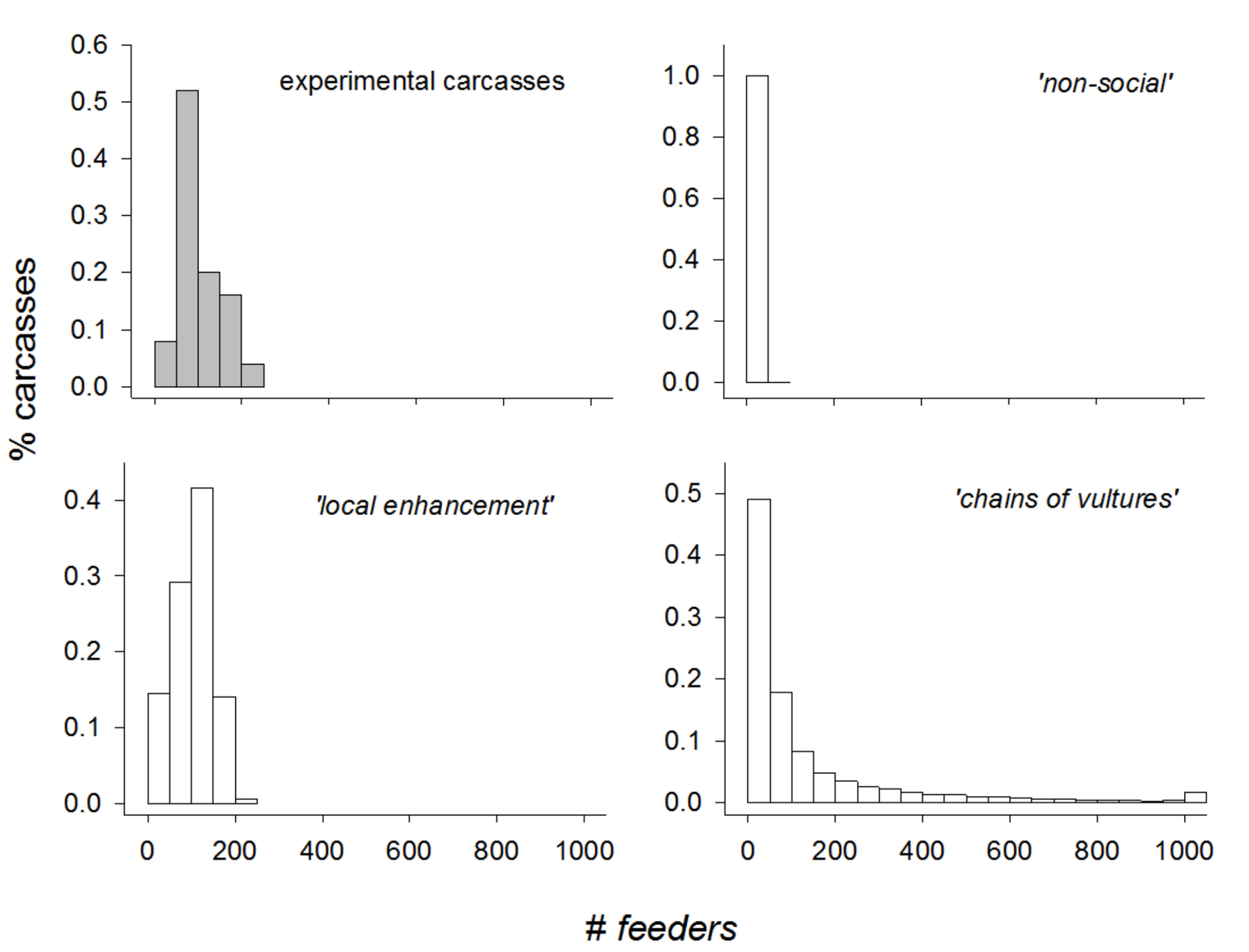



A. Cortes-Avizanda *et al.*, "Bird sky networks: How do avian scavengers use social information to find carrion?" Ecology **95**, 1799–1808 (2014)

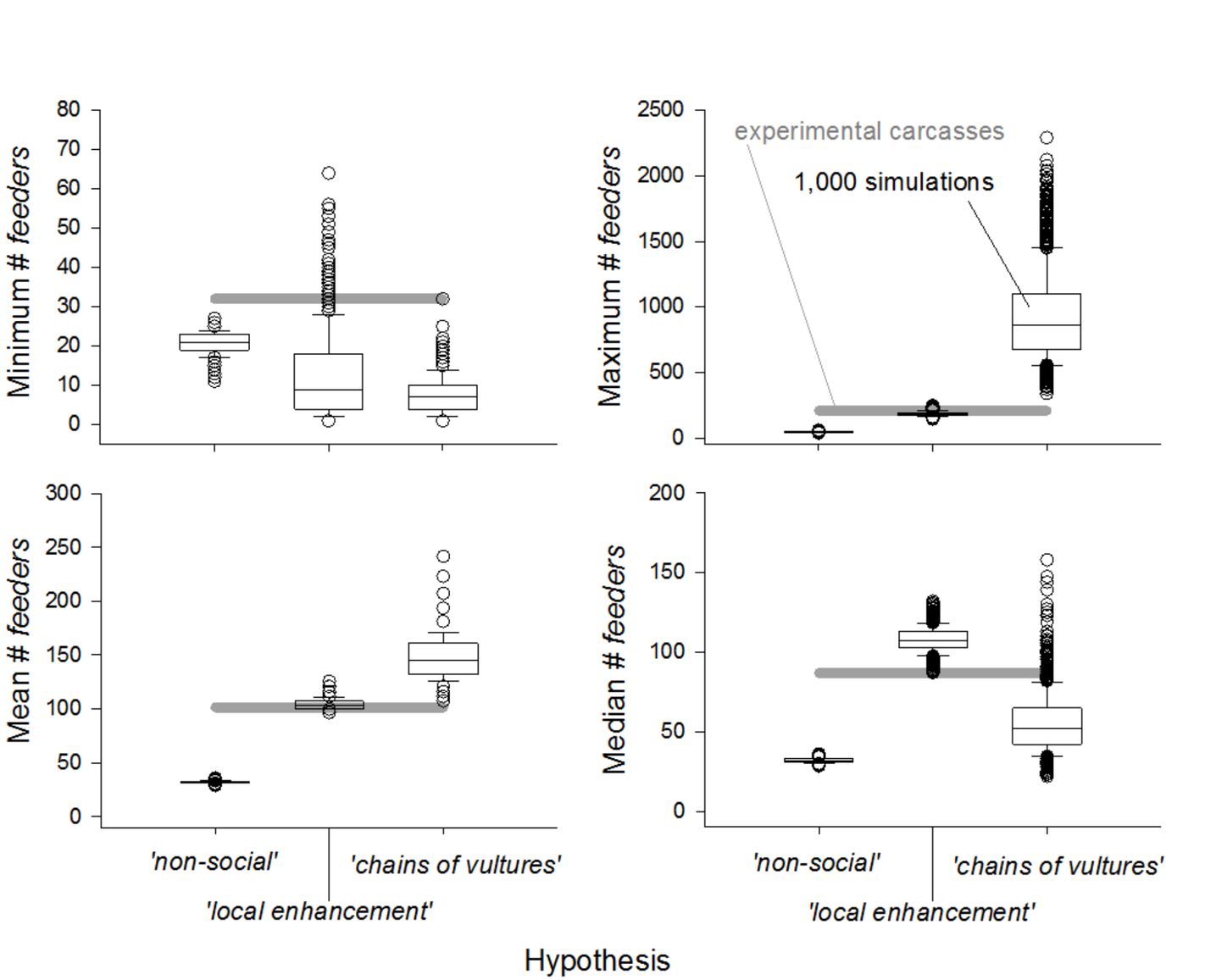
# Interactions among Vultures



# Patterns of Feeding



# Model Comparisons

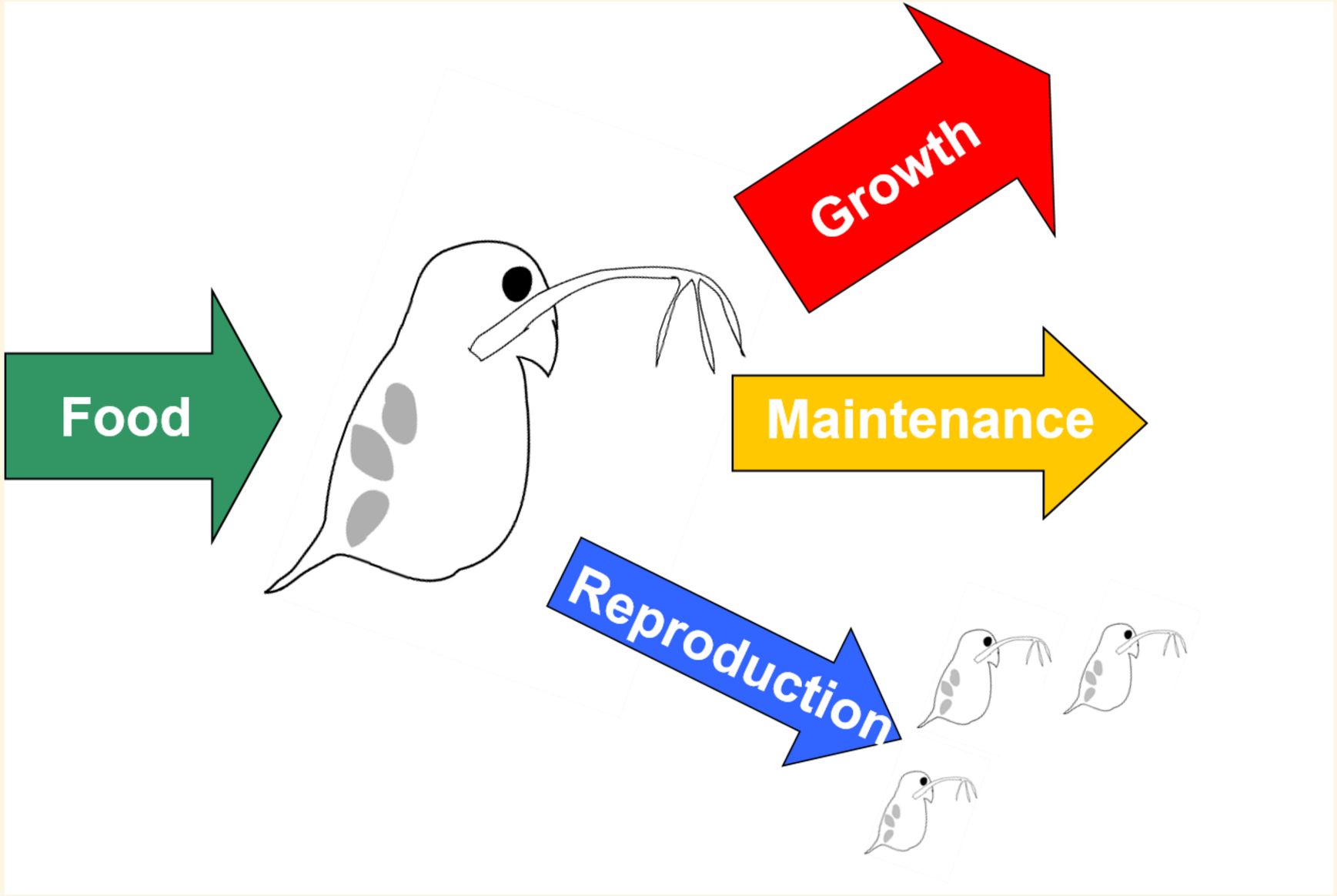


# Impacts of Pesticides on Aquatic Ecosystems

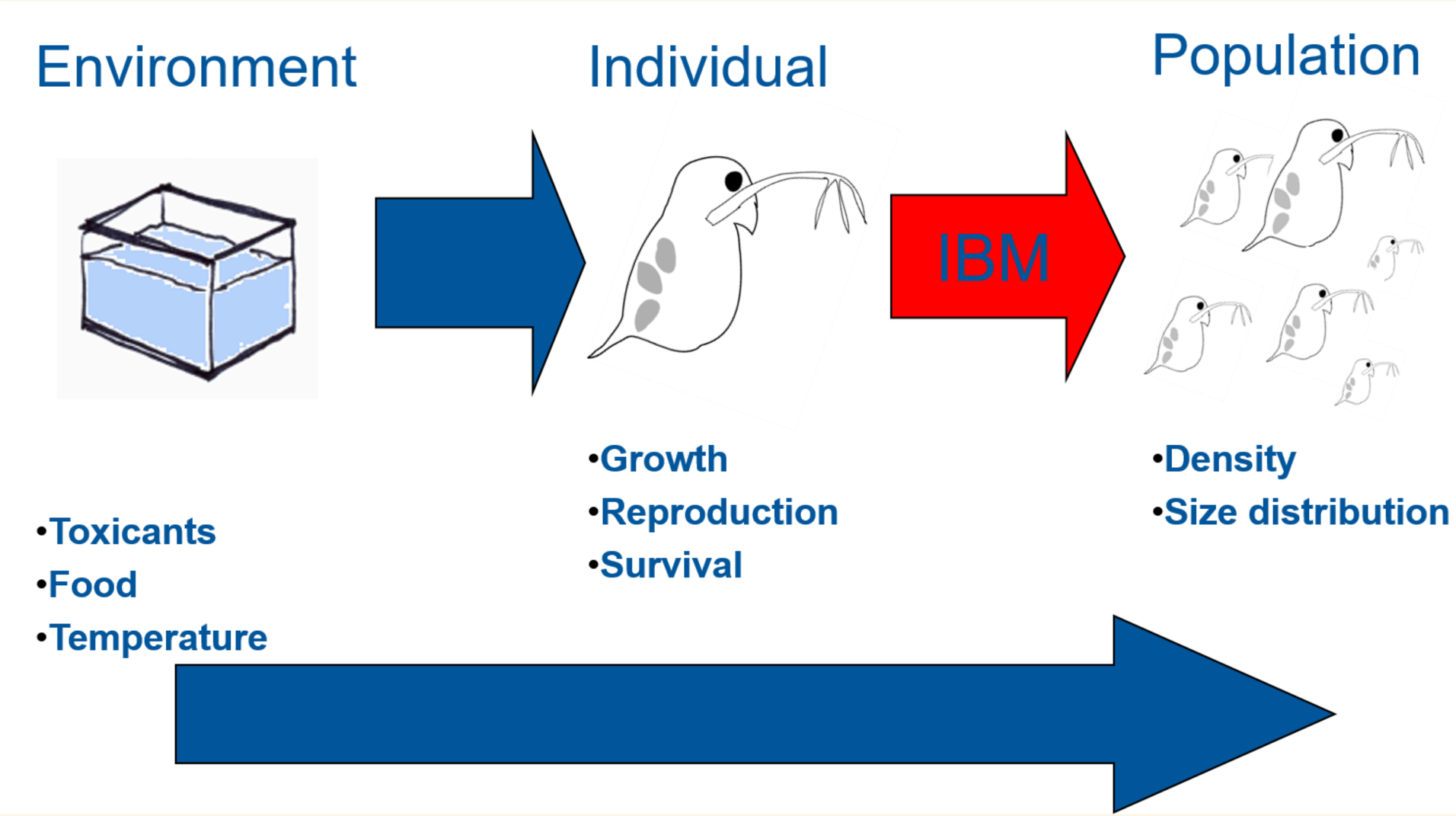
# Dynamic Energy Budget Model

- *Daphnia* (water flea) population dynamics observed in laboratory
- Previous models focused on details of each species
- Dynamic Energy Budget is generic:
  - Calibrate with known species, apply to new (unknown) species

# Individual Metabolic Energy Balance

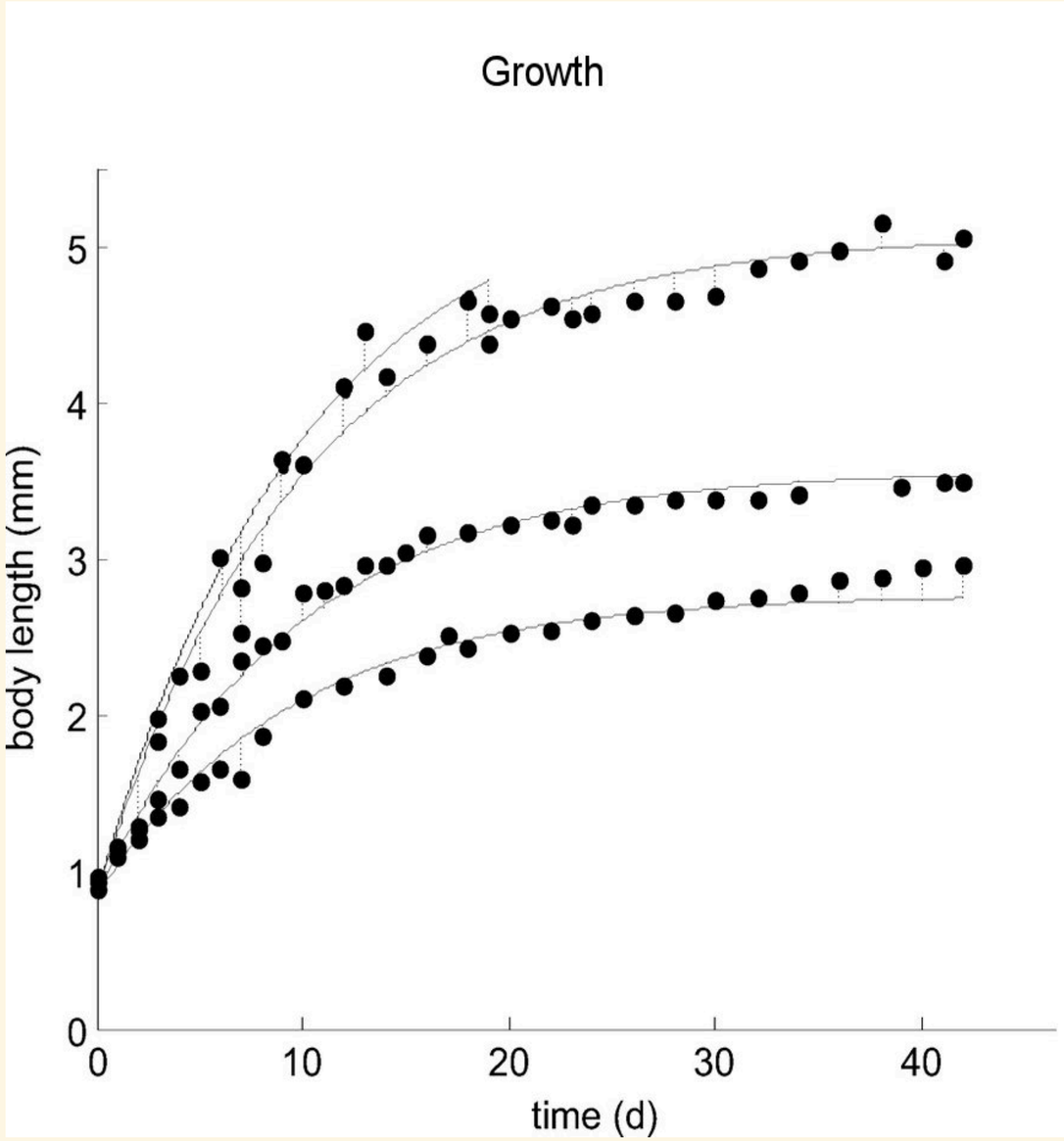


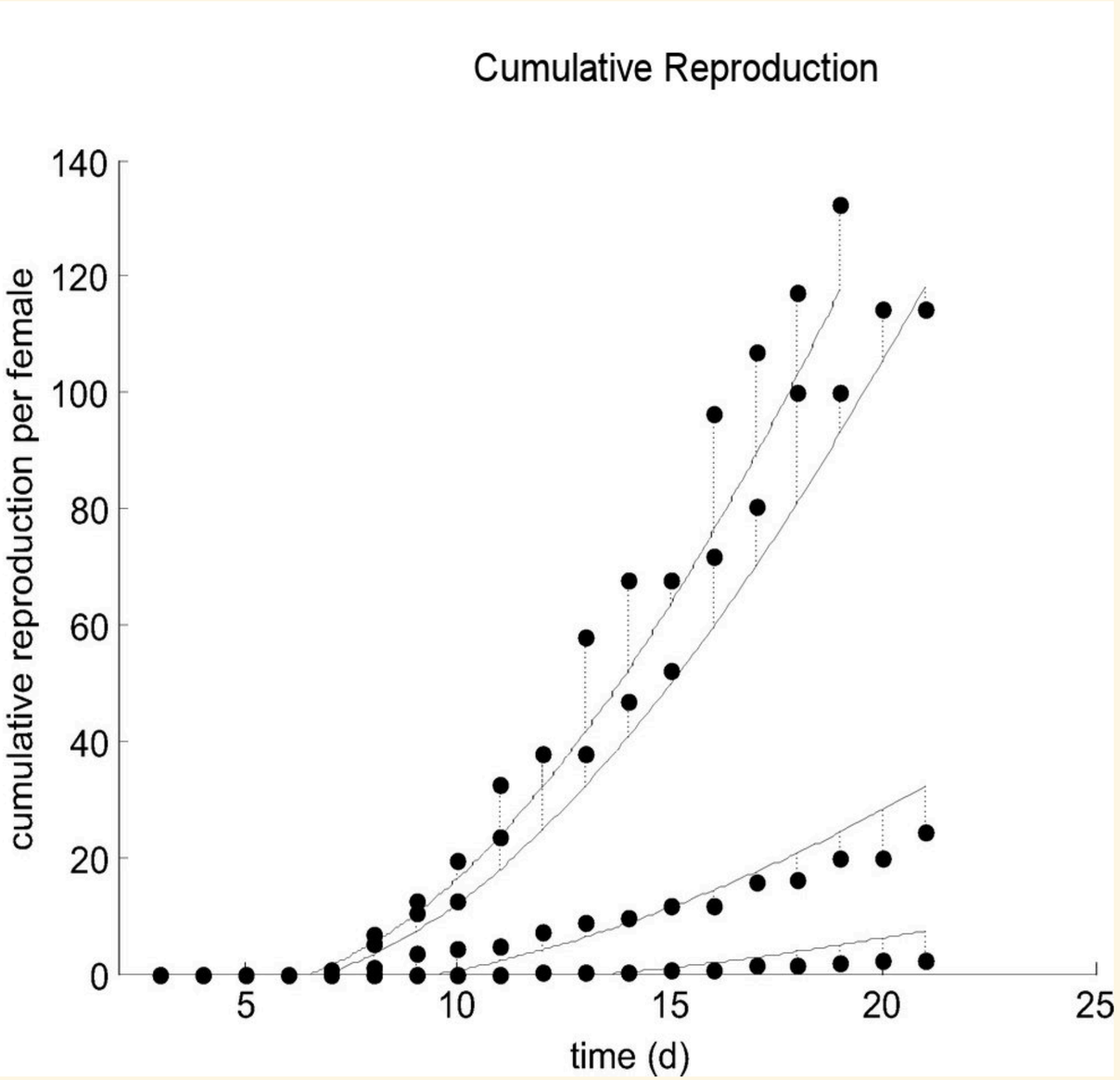
# Dynamic Energy Balance Model in NetLogo




B.T. Martin *et al.*, "Predicting Population Dynamics from the Properties of Individuals: A Cross-Level Test of Dynamic Energy Budget Theory", Am. Naturalist **181**, 506–519 (2013)

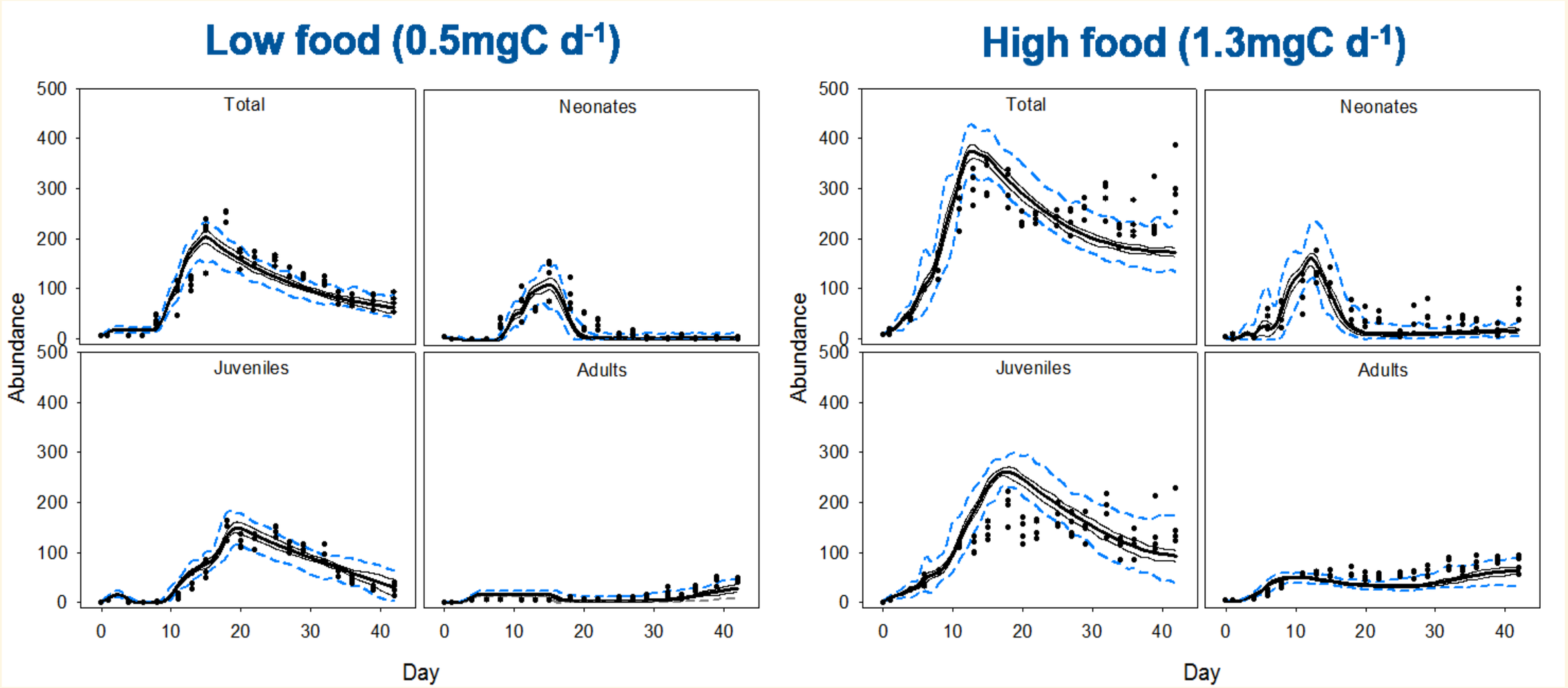
# Testing the Model



# Different Environmental Conditions

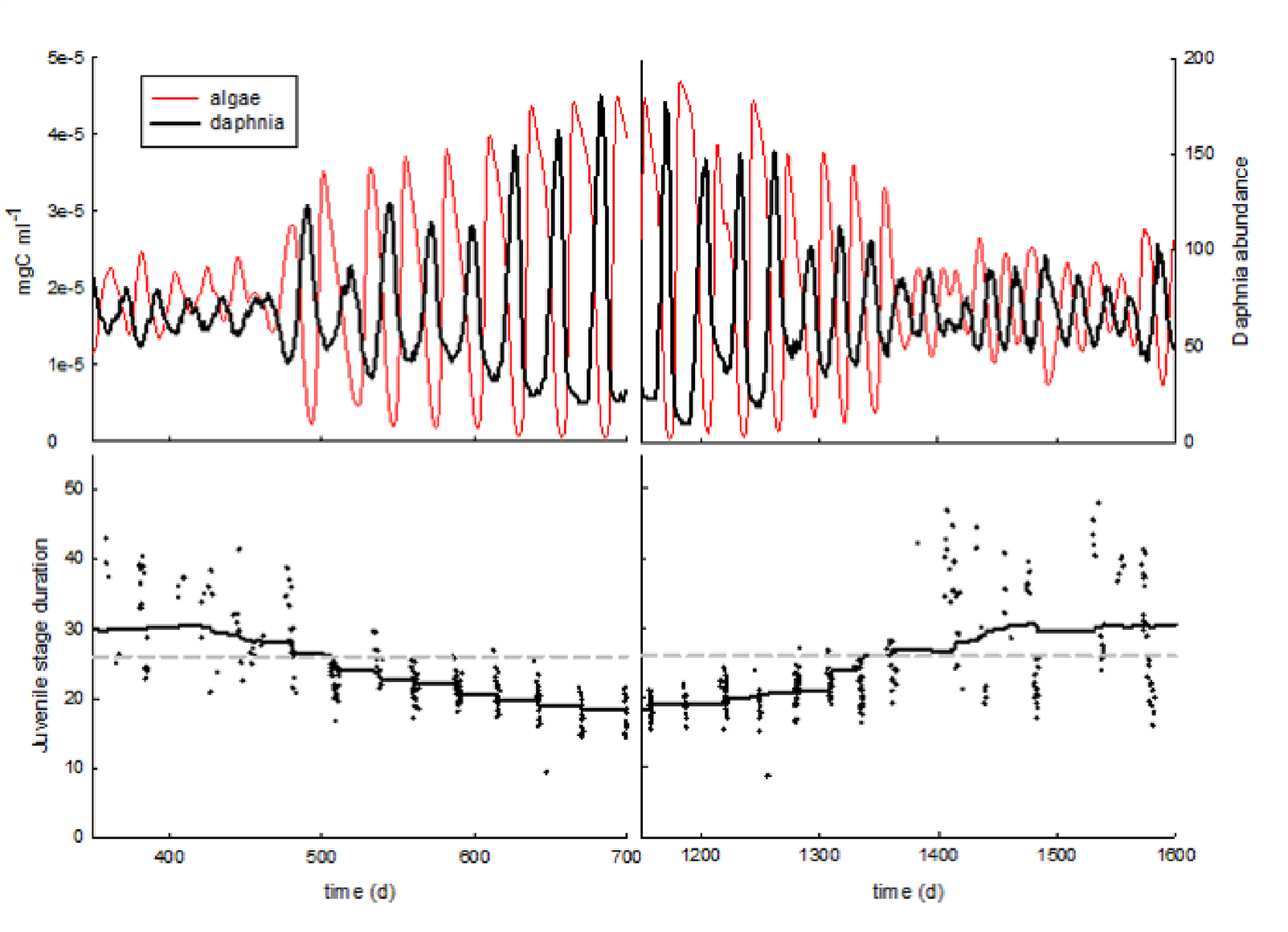
## Model reproduces population density and body size distribution at multiple levels of food supply and toxicant exposure



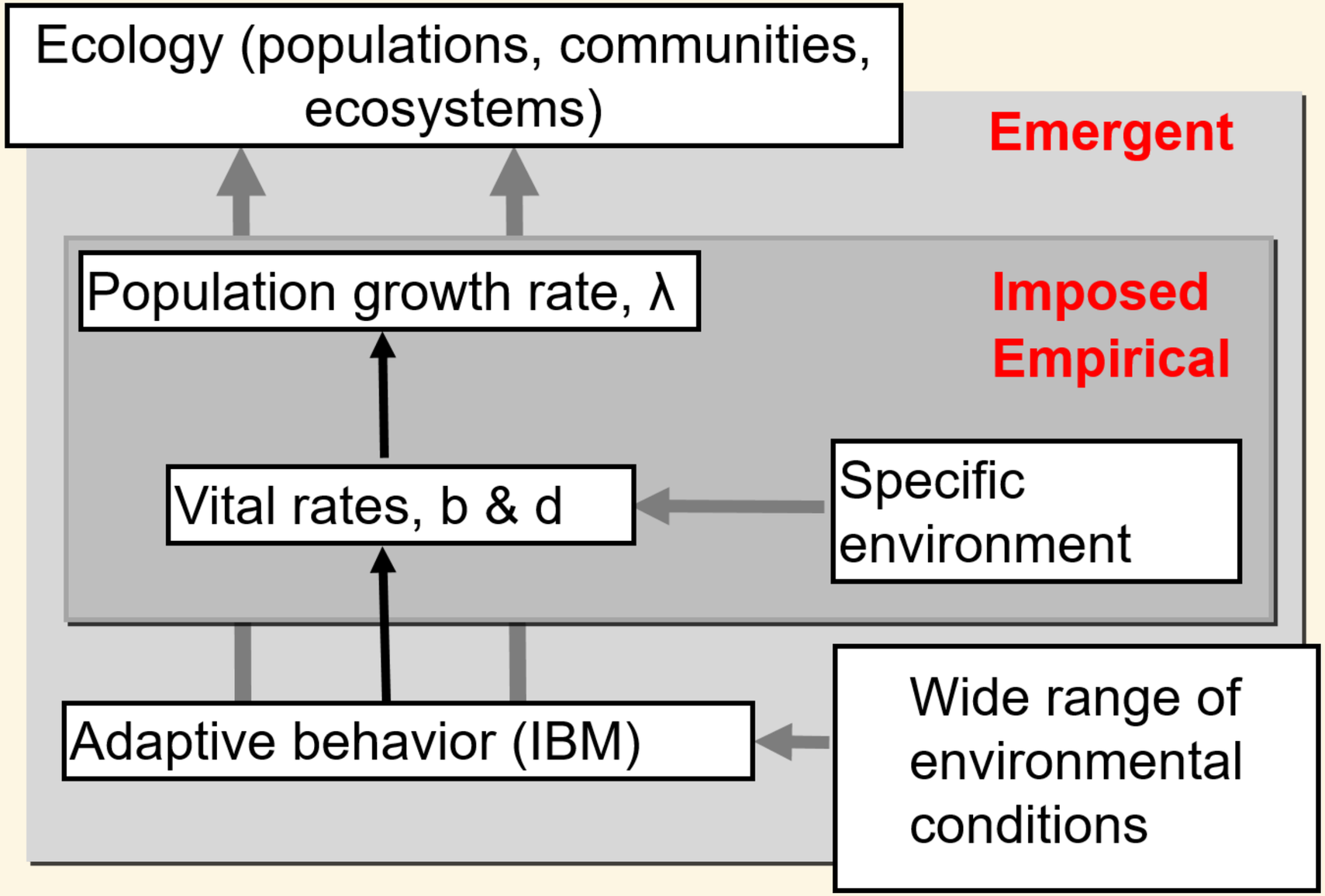
# Dynamical Patterns: Population Cycles

- Experiments find that predator-prey population cycles can occur between *Daphnia* (predator) and algae (prey).
- These can have small amplitude (SA) or large amplitude (LA)
- A pattern is that when the population is the relation of amplitude to the duration of the juvenile stage of *Daphnia* development
  - In SA oscillations, juvenile stage lasts longer than average.
  - In LA oscillations, the juvenile stage is shorter than average.

# Simulations of Population Cycles



# How Model Fits into Research



# Pattern-Oriented Process

# Individual-Based Ecology

*Overview Articles*

2015. BioScience 65: 140-150

## Making Predictions in a Changing World: The Benefits of Individual-Based Ecology

RICHARD A. STILLMAN, STEVEN F. RAILSBACK, JARL GISKE, UTA BERGER, AND VOLKER GRIMM






# Phase-1: Conceptualization

- Define research question(s)
  - Determine whether agent-based/individual-based modeling is the right conceptual framework
- Identify link between research question and behavioral mechanisms
- Identify key parameters and processes to represent environment and behavior of the agents (biological species, human actors, etc.)

# Phase-2: Implementation

- Start with proof-of-concept:
- Program initial model
  - Simple, general model
  - Ideally, starting model should contain little or no code or submodels specific to a single situation
- Test whether model can produce predictions that are *accurate enough* to answer research question
- Run sensitivity analysis of model to determine *key parameters and processes* and relationship between model complexity and predictive power

# Phase-3: Diversification

- Simplify model as much as possible by removing unnecessary parameters and processes
- Minimize number of parameters (e.g., global variables) that need to be measured in each new system
  - Derive as many of these as you can from research literature or general relationships
- Parameterize and test simplified model for a wide range of systems to determine limits of approach
- Perform *meta-analysis* of model runs specific to different sites, or situations
- Use model to test more general theories: gain broadly applicable insights.